

AI Product Advisor for Real Customer Needs

A decision-layer assistant that turns vague shopping needs into a trusted Top 3 shortlist with clear trade-offs.

Retail AI

Vietnamese natural language

Grounded recommendations

THE PROBLEM

Customers do not want more specs. They want confidence.

Electronics shopping forces everyday buyers to translate life context into technical filters: BTU, inverter, noise level, capacity, stock, warranty, delivery and promotion rules

Too many SKU choices

Users compare long lists but still do not know which difference matters.

Spec literacy gap

"Room 18m², quiet, save electricity" must become product constraints.

Low decision trust

FROM COMPARISON TO DECISION

Price, promotion and stock change by region; advice must be

“Help me turn a fuzzy need into a suitable, feasible, easy-to-understand buying decision.”

The job is not “use AI to compare products.” The job is “know which product fits my situation and why.”

Em muốn mua máy lạnh dưới 20 triệu cho phòng ngủ 18m², tiết kiệm điện, ít ồn.

Mình đã hiểu ngân sách, diện tích và ưu tiên. Để chọn đúng hơn, phòng có bị nắng trực tiếp không?

Outcome: fewer choices, clearer reasons, no invented price or stock.

Retail websites already have information, navigation and transaction. The missing layer is decision.

Info

Specs, descriptions, reviews, policies and product pages.

03 • OPPORTUNITY

Nav

Search, category browsing, filters and comparison tables.

Txn

Price, promotion, stock, delivery, installment and checkout.

THE DECISION LAYER

SOLUTION

One AI advisor, three jobs.

The assistant behaves like a careful product consultant, not a generic chatbot.

1

Understand

Extract category, budget, usage context, hard constraints and soft preferences from Vietnamese input.

2

Clarify

Ask only 2–3 questions that materially change filtering or ranking.

3

Recommend

Return Top 3 options with why-fit, weakness, trade-off, source and next action.

From natural language to a grounded recommendation.



Core behavior: ask when uncertainty is important; act when confidence is high;
refuse to invent when data is missing.

ASK • ACT • VERIFY

Start with air conditioners because the recommendation logic is clear and testable.

Why air conditioners?

Need maps naturally to decision rules: room area, sunlight, bedroom use, BTU/HP range, inverter, electricity cost, noise, installation region and stock.

MVP input example

"Tôi cần máy lạnh dưới 20 triệu cho phòng ngủ 18 m², muốn tiết kiệm điện và chạy êm."

The answer should feel like advice, not a spec table.

Best overall

Model A Inverter 1.5HP
18.9M VND

Best balance for 18m² bedroom:
suitable capacity, quiet mode, inverter
savings. Weakness: not the cheapest.

Best budget

Model B 1.5HP
15.7M VND

Good core fit under budget. Trade-
off: slightly higher noise and less
premium energy-saving features.

Quietest

**Model C Premium
Inverter**
19.8M VND

Stronger fit if sleep comfort matters
most. Trade-off: highest price and
may require stock check by region.

07 • EXPERIENCE

Every factual claim must map back to catalog, price, stock, promotion or policy data.

EXPLAIN THE TRADE-OFF

AI where it helps, deterministic logic where it protects trust.

Input guardrail

Scope, language, word limit, unsafe input and basic validation.

Intent + need extractor

Classify category, extract budget, constraints, preferences and missing variables.

State merge

Session memory, confirmed assumptions, prior products and clarification history.

Catalog retrieval

Search product data, price, stock, promotion, review and policy sources.

Hard filter + ranking

Filter by must-have constraints, then rank by fit-to-need and trade-off role.

Grounded answer

Top 3, explanation, sources, uncertainty and next action.

GROUNDING BY DESIGN

Trust is earned through expectation, explainability and control.

The system should never pretend to know what the data does not show.

Expectation

Make clear: this is a shopping advisor, not a guarantee of availability or final pricing.

Explainability

Show why each product is recommended and which input factors mattered.

Control

Let users change priorities, relax constraints, see sources, or hand off to a human consultant.

EVALUATION

Quality gate: measure the AI task, not just product usage.

$\geq 90\%$

Need extraction accuracy for category, budget, room size and priority.

0

Critical hallucinations on price, promotion, stock or specs.

$< 5s$

Top 3 comparison response time on demo data.



Code checks

Schema, enum, source IDs, price bounds, latency and forbidden claims.

10 • EVALS



Human review

APPLICATION-LEVEL QUALITY

PILOT PATHWAY

Three-month pilot on one website/app or one product category.

1K–10K

Trial conversations with anonymized/demo customer contexts.

3 mo

Post-hackathon pilot with catalog, price, stock and policy data.

KPI

Correctness, no serious hallucination, usability, traceability and API readiness.

PILOT Partner support: cleaned/anonymized product data, domain expert review, test environment and business feedback

FROM DEMO TO DEPLOYMENT

Ship a believable slice, not a full retail platform.

0–6h

Dataset + scenarios

Prepare catalog sample, policy snippets and 20–30 customer need cases.

6–18h

Core agent

Build extraction, clarification, deterministic filtering and ranker.

18–34h

Web demo

Implement chat UI, Top 3 comparison, source display and fallback states.

34–48h

Evals + pitch

Run test suite, log traces, polish demo story and pilot plan.

CLOSING

The winning product is not a chatbot. It is a trustworthy decision engine for retail.

Start with air conditioners, prove decision quality, then expand across high-consideration electronics categories.

Build

Demo-ready AI advisor with Top 3 recommendations.

Measure

Groundedness, need understanding, trade-off quality and conversion signals.

Scale

Category-by-category rollout with trace-driven eval flywheel.

THANK YOU

AI PRODUCT ADVISOR